

YANG YU

Research Associate III
Physical Oceanography
Woods Hole Oceanographic Institution

Phone: (530) 220-3253
E-mail: yang.yu@whoi.edu
ORCID: 0000-0002-5527-1540

PROFESSIONAL EXPERIENCE:

2026.05-present Research Associate III, Woods Hole Oceanographic Institution
2025.05-2026.04 Postdoctoral Investigator, Woods Hole Oceanographic Institution
2022.12-2025.04 Postdoctoral Associate, University of California, Davis
2021.08-2022.11 Postdoctoral Scholar, Institute of Oceanology Chinese Academy of Sciences

EDUCATION:

2016-2021 Ph.D., Environmental Science, Ocean University of China
2014-2016 M.Sc., Environmental Engineering, Ocean University of China
2010-2014 B.Sc., Environmental Science, Ocean University of China

RESEARCH ACTIVITIES:

- **Impacts of Offshore Wind Energy on the Northeast U.S. Continental Shelf (2025-)**
I am currently developing a coupled atmosphere-ocean-biogeochemistry model based on WRF, ROMS, and Darwin to investigate the impacts of offshore wind farms on ocean currents, local mixing, and marine biogeochemistry on the Northeast U.S. Continental Shelf.
- **Agricultural Fallowing, Dust Emissions, and Air-Quality Hazards (2024-)**
I use WRF-Chem and high-resolution land-use data to investigate how agricultural fallowing and drought-driven land-surface changes influence dust emissions, visibility degradation, and air-quality hazards in California's Central Valley.
- **Agricultural Irrigation Impacts on Local Meteorology and Air Quality (2022-)**
I developed a WRF-irrigation model to examine the impacts of agricultural irrigation on surface meteorology and air quality in California's Central Valley.
This WRF-irrigation model serves as the next-generation baseline forecasting model in the California Air Resources Board (CARB) air quality forecasting system.
- **Hybrid Atmosphere-Ocean Coupled Modeling (2021-)**
I developed a hybrid coupled model based on the Regional Ocean Modeling System (ROMS) and a statistical atmospheric model to investigate multi-scale atmosphere-ocean interactions and biogeochemical feedbacks associated with ENSO in the tropical Pacific.
- **High-Resolution Atmosphere-Ocean Coupled Modeling (2018-2022)**
I investigated eddy-mean flow interactions during the rapid intensification of Super Typhoon Muifa (2011) using an eddy-resolving-resolution (1 km) WRF-ROMS coupled model. I also studied hurricane-induced remote extreme precipitation using a cloud-permitting-resolution (4 km) WRF-ROMS coupled model.
- **Diurnal Effects on Marginal Seas (2016-2020)**
I developed a three-dimensional regional circulation model based on the Regional Ocean Modeling System (ROMS) to investigate seasonal variability in water temperature and salinity driven by diurnal surface forcing in the East China Sea.

TEACHING EXPERIENCE:

Teaching Assistant, Numerical Modeling of the Atmosphere, UC Davis - Winter 2025
Teaching Assistant, Numerical Modeling of the Atmosphere, UC Davis - Winter 2023

OUTREACH AND ACTIVITIES

Seminar Organizer, Department of Physical Oceanography, Woods Hole Oceanographic Institution, Woods Hole, MA, U.S., September 2025-present

HONORS AND AWARDS

China Scholarship Council (CSC) Scholarship, 2018

National Scholarship for Graduate Students, China, 2018

Outstanding Graduate Student, Ocean University of China, 2015

National Scholarship for Graduate Students, China, 2015

PUBLICATIONS

• Peer-reviewed Journal Publications:

2026

1. Yang, Y., Gao, T., Xu, Y., **Yu, Y.**, Liu, X., Xie, L., Zhang, W., Zhang, B., Guo, X., Long, X., & Zhu, G. (2026). Increasing Socioeconomic Exposure to Extreme Tropical Cyclone Precipitation over China. *Journal of Hydrometeorology*, 27(6), 937–949. <https://doi.org/https://doi.org/10.1175/JHM-D-25-0216.1>
2. Lee, A., Chen, S.-H., Huang, C.-C., Nathan, T. R., **Yu, Y.**, Kavaya, M. J., Bedka, K. M., Nehrir, A. R., Thornhill, K. L., Barton-Grimley, R. A., Collins, J. E., Cooney, J. W., Ferrare, R. A., Liu, Z., Lambriksen, B., Schreier, M., Wong, S., & Veals, P. G. (2026). Improved Analysis and Prediction of Tropical Storm Hermine (2022) over the Eastern Tropical Atlantic Using CPEX-CV Observations. *Monthly Weather Review*. <https://doi.org/10.1175/MWR-D-25-0062.1>

2025

3. Li, Y., **Yu, Y.**, Gao, C., Wang, H., & Zhang, R.-H. (2025). Tropical instability waves and their relationships with large-scale oceanic thermal conditions, revealed in ROMS-based hybrid coupled model of the tropical Pacific. *Journal of Oceanology and Limnology*. <https://doi.org/10.1007/s00343-025-5053-9>
4. **Yu, Y.**, Li, Y.-N., Zhang, R.-H., Chen, S.-H., Tseng, Y.-H., Zhang, W., & Wang, H. (2025). A flexible Regional Ocean Modeling System-based hybrid coupled model for El Niño–Southern Oscillation studies – model formulation and performance evaluation. *Geoscientific Model Development*, 18(17), 5527–5547. <https://doi.org/10.5194/gmd-18-5527-2025>
5. **Yu, Y.**, Chen, S.-H., Huang, C.-C., Paw U, K. T., Schmitt, C., Zhao, Z., Suvočarev, K., Kisekka, I., Pyles, R. D., Avise, J., Cai, C., Chang, K.-Y., & Lo, M.-H. (2025). A Numerical Study of the Agricultural Irrigation Effects on Summer Soil Moisture and Near-Surface Meteorology in California’s Central Valley. *Journal of Hydrometeorology*, 26(6), 641–659. <https://doi.org/10.1175/JHM-D-24-0129.1>
6. Gong, X., **Yu, Y.**, Shi, X., Lin, X., Liu, G., Dong, Z., Wang, X., Zheng, J., Lembke-Jene, L., & Lohmann, G. (2025). Thresholds in East Asian marginal seas circulation due to deglacial sea level rise. *Npj Climate and Atmospheric Science*, 8(1), 83. <https://doi.org/10.1038/s41612-025-00927-y>
7. Zhang, R.-H., Zhang, W., **Yu, Y.**, Li, Y., Tian, F., Gao, C., & Wang, H. (2025). A hybrid coupled model for the tropical Pacific constructed by integrating ROMS with a statistical atmospheric model. *Journal of Oceanology and Limnology*, 43(4), 1037–1055. <https://doi.org/10.1007/s00343-024-4194-6>

2024

8. Zhang, W., Gao, C., Tian, F., **Yu, Y.**, Wang, H., & Zhang, R.-H. (2024). Representing ocean biology-induced heating effects in ROMS-based simulations for the Indo-Pacific Ocean. *Frontiers in Marine Science*, 11(November), 1–17. <https://doi.org/10.3389/fmars.2024.1473208>

2023

9. Geng, X., Wu, C., Yang, Z., Zhu, J., Tang, K., Lin, J., Liu, Y., Zhang, Y., An, M., Zhao, W., & **Yu, Y.** (2023). Spatiotemporal variations and impact factors of nutrients in the Sanya Bay, northern South China Sea. *Environmental Science and Pollution Research*, 30(31), 76784–76797. <https://doi.org/10.1007/s11356-023-27527-8>
10. Kuo, Y.-C., **Yu, Y.**, & Tseng, Y.-H. (2023). Interannual changes of the summer circulation and hydrology in the East China Sea: A modeling study from 1981 to 2015. *Ocean Modelling*, 181(November 2022), 102156. <https://doi.org/10.1016/j.ocemod.2022.102156>
11. Zhu, J., Zhou, Q., Zhou, Q., Geng, X., Shi, J., Guo, X., **Yu, Y.**, Yang, Z., & Fan, R. (2023). Interannual variation of coastal upwelling around Hainan Island. *Frontiers in Marine Science*, 10(January), 1–10. <https://doi.org/10.3389/fmars.2023.1054669>

2022

12. Liu, H., Lin, L., Wang, Y., Du, L., Wang, S., Zhou, P., **Yu, Y.**, Gong, X., & Lu, X. (2022). Reconstruction of Monthly Surface Nutrient Concentrations in the Yellow and Bohai Seas from 2003–2019 Using Machine Learning. *Remote Sensing*, 14(19), 5021. <https://doi.org/10.3390/rs14195021>
13. **Yu, Y.**, Chen, S., Tseng, Y., Foltz, G. R., Zhang, R., & Gao, H. (2022). Impacts of model resolution and ocean coupling on mean and eddy momentum transfer during the rapid intensification of super-typhoon Muifa (2011). *Quarterly Journal of the Royal Meteorological Society*, 148(749), 3639–3659. <https://doi.org/10.1002/qj.4379>
14. **Yu, Y.**, Gao, T., Xie, L., Zhang, R.-H., Zhang, W., Xu, H., et al. (2022). Tropical cyclone over the western Pacific triggers the record-breaking ‘21/7’ extreme rainfall in Henan, central-eastern China. *Environmental Research Letters*, 17(12), 124003. <https://doi.org/10.1088/1748-9326/aca2c4>

2021

15. Meng, S., Gong, X., **Yu, Y.**, Yao, X., Gong, X., Lu, K., et al. (2021). Strengthened ocean-desert process in the North Pacific over the past two decades. *Environmental Research Letters*. <https://doi.org/10.1088/1748-9326/abd96f>
16. Zhang, C., Shi, Z., Zhao, J., Zhang, Y., **Yu, Y.**, Mu, Y., et al. (2021). Impact of air emissions from shipping on marine phytoplankton growth. *Science of The Total Environment*, 769, 145488. <https://doi.org/10.1016/j.scitotenv.2021.145488>

2020

17. Ding, X., Guo, X., Zhang, C., Yao, X., Liu, S., Shi, J., Luo, C., Yu, X., **Yu, Y.**, & Gao, H. (2020). Water conservancy project on the Yellow River modifies the seasonal variation of Chlorophyll-a in the Bohai Sea. *Chemosphere*, 254, 126846. <https://doi.org/10.1016/j.chemosphere.2020.126846>
18. Feng, L., **Yu, Y.**, Gao, H., & Yao, X. (2020). Surface water formation on the natural surface under supersaturation: from local water balance to air pollutant deposition. *Air Quality, Atmosphere & Health*. <https://doi.org/10.1007/s11869-020-00901-y>
19. Qi, J., **Yu, Y.**, Yao, X., Gang, Y., & Gao, H. (2020). Dry deposition fluxes of inorganic nitrogen and phosphorus in atmospheric aerosols over the Marginal Seas and Northwest Pacific. *Atmospheric Research*, 245(April), 105076. <https://doi.org/10.1016/j.atmosres.2020.105076>
20. **Yu, Y.**, Chen, S.-H., Tseng, Y.-H., Guo, X., Shi, J., Liu, G., Zhang, C., Xu, Y., & Gao, H. (2020). Importance of Diurnal Forcing on the Summer Salinity Variability in the East China Sea. *Journal of Physical Oceanography*, 50(3), 633–653. <https://doi.org/10.1175/JPO-D-19-0200.1>
21. Zhang, C., He, J., Yao, X., Mu, Y., Guo, X., Ding, X., **Yu, Y.**, Shi, J., & Gao, H. (2020). Dynamics of phytoplankton and nutrient uptake following dust additions in the northwest Pacific. *Science of The Total Environment*, 739, 139999. <https://doi.org/10.1016/j.scitotenv.2020.139999>

Before 2020

22. Liu, Q., Jiang, Y., Wang, Q., Wang, M., McMinn, A., Zhao, Q., Xue, C., Wang, X., Dong, J., **Yu, Y.**, Han, Y., & Zhao, J. (2019). Using picoeukaryote communities to indicate the spatial heterogeneity of the Nordic Seas. *Ecological Indicators*, 107, 105582. <https://doi.org/10.1016/j.ecolind.2019.105582>

23. Zhang, C., Yao, X., Chen, Y., Chu, Q., **Yu, Y.**, Shi, J., & Gao, H. (2019). Variations in the phytoplankton community due to dust additions in eutrophication, LNLC and HNLC oceanic zones. *Science of The Total Environment*, 669, 282–293. <https://doi.org/10.1016/j.scitotenv.2019.02.068>
 24. Zhu, Y., Li, K., Shen, Y., Gao, Y., Liu, X., **Yu, Y.**, Gao, H., & Yao, X. (2019). New particle formation in the marine atmosphere during seven cruise campaigns. *Atmospheric Chemistry and Physics*, 19(1), 89–113. <https://doi.org/10.5194/acp-19-89-2019>
 25. Zhang, C., Gao, H., Yao, X., Shi, Z., Shi, J., **Yu, Y.**, Meng, L., & Guo, X. (2018). Phytoplankton growth response to Asian dust addition in the northwest Pacific Ocean versus the Yellow Sea. *Biogeosciences*, 15(3), 749–765. <https://doi.org/10.5194/bg-15-749-2018>
 26. Xu, Z., Song, X., Wang, M., Liu, Q., Jiang, Y., Shao, H., Liu, H., Shi, K., & **Yu, Y.** (2017). Community patterns and temporal variation of picoeukaryotes in response to changes in the Yellow Sea Warm Current. *Journal of Oceanography*, 73(5), 687–699. <https://doi.org/10.1007/s10872-017-0425-1>
 27. **Yu, Y.**, Gao, H., Shi, J., Guo, X., & Liu, G. (2017). Diurnal Forcing Induces Variations in Seasonal Temperature and Its Rectification Mechanism in the Eastern Shelf Seas of China. *Journal of Geophysical Research: Oceans*, 122(12), 9870–9888. <https://doi.org/10.1002/2017JC013473>
 28. Meng, L., Gao, H. W., **Yu, Y.**, Yao, X. H., Gao, Y., Zhang, C., & Fan, L. (2017). A new approach developed to study variability in North African dust transport routes over the Atlantic during 2001–2015. *Geophysical Research Letters*, 44(19), 10,026–10,035. <https://doi.org/10.1002/2017GL074478>
 29. **Yu, Y.**, Gao, H., & Shi, J. (2017). Impacts of Diurnal Forcing on Temperature Simulation in the Shelf Seas of China. *Periodical of Ocean University of China*, 47(4), 106–113. <https://doi.org/10.16441/j.cnki.hdxh.20160138>
 30. **Yu, Y.**, Sun, P., Cui, H., Sheng, H., Zhao, F., Tang, Y., & Chen, Z. (2015). Effects of trawl selectivity and genetic parameters on fish body length under long-term trawling. *Journal of Ocean University of China*, 14(5), 835–840. <https://doi.org/10.1007/s11802-015-2885-5>
 31. Sun, P., Liang, Z.-L., **Yu, Y.**, Tang, Y.-L., Zhao, F.-F., & Huang, L.-Y. (2015). Trawl selectivity-induced evolution effects on age structure and size-at-age of largehead hairtail (*Trichiurus lepturus*) Linnaeus, 1758 in the East China Sea, China. *Journal of Applied Ichthyology*, 31(4), 657–664. <https://doi.org/10.1111/jai.12774>
- **Submitted:**
 1. **Yu, Y.**, Chen, S.-H., Adebisi, A., Kibria, M. M., & Pandey, S. (2026). Agricultural Fallowing Drives Extreme Anthropogenic Dust and Visibility Degradation During the October 2021 Dust Event in California's Central Valley. Submitted to *Geophysical Research Letters* <https://doi.org/10.31223/X50195>
 2. Gao, T., **Yu, Y.**, Chen, S.-H., Zhang, W., Gao, J., Liu, J., Major contribution of the western Pacific subtropical high to tropical cyclone remote precipitation extremes during the “21.7” Henan event, Submitted to *Journal of Geophysical Research - Atmospheres*.
 3. Chen, S.-H., Yang, S.-C., Huang, C.-C., **Yu, Y.**, Matsui, T., Zhu, Y., Fu, B. Xue, X. Chen, C.-Y., Influence of aerosol-attenuated Backscatter Data Assimilation on Dust Analysis and Forecasts over the Tropical Atlantic Ocean, Submitted to *Quarterly Journal of the Royal Meteorological Society*.
 4. Wang, K.-Y., Chen, S.-H., Schmitt, C., **Yu, Y.**, Suvocarev, K., McDonald, I., Chen, X.-M., Di, P., Liu, K., Hus, Y.-K., Thiruvengatachari, R., Avise, J., Impacts of Marine Air Penetration and Irrigation on Planetary Boundary Layer Height in California's Central Valley, Submitted to *Journal of Geophysical Research - Atmospheres*.
 - **In Preparation:**
 1. **Yu, Y.**, et al. A flexible Regional Ocean Modeling System-based hybrid coupled model for El Niño–Southern Oscillation studies. Part II: freshwater flux effects.
 2. **Yu, Y.**, et al. A flexible Regional Ocean Modeling System-based hybrid coupled model for El Niño–Southern Oscillation studies. Part III: biogeochemical feedback.